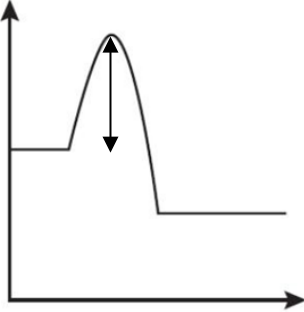


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)			Advantages award (1) each for up to two of following - cane sugar is a renewable raw material / sustainable - plant photosynthesis uses CO₂ and fermentation produces CO₂ making it carbon neutral - low pressure making it cheaper to operate / safer to operate Disadvantage award (1) each for up to two of following - dilute solution of ethanol formed / ethanol not pure - therefore needs further processing / distillation - carbon dioxide formed - contributes to global warming - batch process so labour intensive / inefficient / time consuming - crops used therefore less land for food production / more expensive food	2		2	4		
	(b)			$\text{C}_2\text{H}_5\text{OH} + \text{O}_2 \rightarrow \text{CH}_3\text{COOH} + \text{H}_2\text{O}$ both products needed		1		1		
	(c)	(i)		498 (2) if answer incorrect award (1) for either of following 5616 – (2 × 2061) 1494		2		2	2	
		(ii)		6932 (2) if answer incorrect award (1) for (4 × 805) + (8 × 464)		2		2	2	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iii)		award (1) for any of following - energy released > energy needed - energy out > energy in - overall energy change has a negative value - overall energy change is = -1316		1		1		
		(iv)			1			1		
	(d)			butan-1-ol C butan-2-ol B 2-methylpropan-1-ol D 2-methylpropan-2-ol A award (2) for all four correct award (1) for any two correct	2			2		
				Question 8 total	5	6	2	13	4	0